



**Master of Computer Application – Game Development
Semester – I**

Sub: Front-End Frameworks and Technologies

Topic: Interactive Student Result Dashboard

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INTRODUCTION

The Interactive Student Result Dashboard is a client-side web application developed using HTML5, CSS3, and JavaScript to calculate and display student academic results. It allows users to enter student details and subject-wise marks, and automatically calculates total marks, percentage, grade, and pass/fail status. The application uses JavaScript functions, arrays, objects, conditional statements, DOM manipulation, and event handling to provide dynamic result generation. It also includes student search, Pass/Fail filtering, and clear functionality for better interaction.

Overall, this project demonstrates the practical use of JavaScript-based data processing, dynamic content rendering, and interactive web development in creating a simple student result management system.

INPUT CODE

```
<!doctype html>
<html>
<head>
<title>Student Result</title>
<style>
  body {
    font-family: Arial;
    background: #f2f2f2;
  }
  .box {
    max-width: 500px;
    margin: auto;
    background: white;
    padding: 20px;
  }
  input, select, button {
    padding: 8px;
  }
  input, select {
    width: 95%;
    margin: 5px 0 12px;
  }
  button {
    background: #06c;
    color: white;
    border: 0;
    margin: 5px 0 12px;
  }
  table {
    width: 100%;
    border-collapse: collapse;
  }
```

```

th,
td {
    border: 1px solid #aaa;
    padding: 8px;
}
</style>
</head>
<body>
<div class="box">
    <h1>Student Result</h1>

    <label>Search Student</label>
    <input id="search" type="text" placeholder="Enter name or roll number" />
    <button onclick="searchStudent()">Search</button><br>

    <label>Student Name</label>
    <input id="name" type="text" placeholder="Enter name" />

    <label>Roll Number</label>
    <input id="roll" type="text" placeholder="Enter roll number" />

    <h3>Enter Marks</h3>
    <input id="html" type="number" placeholder="HTML marks" />
    <input id="css" type="number" placeholder="CSS marks" />
    <input id="javascript" type="number" placeholder="JavaScript marks" />

    <label>Filter Result</label>
    <select id="filter">
        <option value="All">Show All</option>
        <option value="Pass">Pass Only</option>
        <option value="Fail">Fail Only</option>
    </select>

    <br />
    <button onclick="showResult()">Show Result</button>
    <button onclick="clearDashboard()">Clear</button>

```

```
<div id="result"></div>
</div>
```

```
<script>
const savedStudents = [
  {
    name: "Aarav Sharma",
    roll: "ST101",
    resultHTML: `
      <h2>Result</h2>
      <p><b>Name:</b> Aarav Sharma</p>
      <p><b>Roll Number:</b> ST101</p>
      <table>
        <tr><th>Subject</th><th>Marks</th></tr>
        <tr><td>HTML</td><td>85</td></tr>
        <tr><td>CSS</td><td>78</td></tr>
        <tr><td>JavaScript</td><td>90</td></tr>
      </table>
      <p><b>Total:</b> 253/300</p>
      <p><b>Percentage:</b> 84.33%</p>
      <p><b>Grade:</b> A</p>
      <p><b>Result:</b> Pass</p>
    `,
  },
  {
    name: "Diya Patel",
    roll: "ST102",
    resultHTML: `
      <h2>Result</h2>
      <p><b>Name:</b> Diya Patel</p>
      <p><b>Roll Number:</b> ST102</p>
      <table>
        <tr><th>Subject</th><th>Marks</th></tr>
        <tr><td>HTML</td><td>65</td></tr>
        <tr><td>CSS</td><td>72</td></tr>
        <tr><td>JavaScript</td><td>68</td></tr>
      </table>
    `
  }
]
```

```

    <p><b>Total:</b> 205/300</p>
    <p><b>Percentage:</b> 68.33%</p>
    <p><b>Grade:</b> B</p>
    <p><b>Result:</b> Pass</p>
    ,
  }
];

function showResult() {
  const name = document.getElementById("name").value;
  const roll = document.getElementById("roll").value;
  const html = Number(document.getElementById("html").value);
  const css = Number(document.getElementById("css").value);
  const javascript = Number(document.getElementById("javascript").value);
  const total = html + css + javascript;
  const percentage = total / 3;
  const status =
    html >= 40 && css >= 40 && javascript >= 40 ? "Pass" : "Fail";
  let grade = "F";

  if (status === "Pass") {
    if (percentage >= 90) grade = "A+";
    else if (percentage >= 75) grade = "A";
    else if (percentage >= 60) grade = "B";
    else if (percentage >= 50) grade = "C";
    else grade = "D";
  }

  const filter = document.getElementById("filter").value;

  if (filter !== "All" && filter !== status) {
    document.getElementById("result").innerHTML =
      "<p>No result matches this filter.</p>";
    return;
  }

  const resultHTML = `

```

```

<h2>Result</h2>
<p><b>Name:</b> ${name}</p>
<p><b>Roll Number:</b> ${roll}</p>
<table>
  <tr><th>Subject</th><th>Marks</th></tr>
  <tr><td>HTML</td><td>${html}</td></tr>
  <tr><td>CSS</td><td>${css}</td></tr>
  <tr><td>JavaScript</td><td>${javascript}</td></tr>
</table>
<p><b>Total:</b> ${total}/300</p>
<p><b>Percentage:</b> ${percentage.toFixed(2)}%</p>
<p><b>Grade:</b> ${grade}</p>
<p><b>Result:</b> ${status}</p>
`;

savedStudents.push( { name, roll, resultHTML } );
document.getElementById("result").innerHTML = resultHTML;
}

function searchStudent() {
  const search = document.getElementById("search").value.toLowerCase();
  const student = savedStudents.find((item) =>
    item.name.toLowerCase().includes(search) ||
    item.roll.toLowerCase().includes(search)
  );

  document.getElementById("result").innerHTML = student
    ? student.resultHTML
    : "<p>Student not found.</p>";
}

function clearDashboard() {
  document.getElementById("search").value = "";
  document.getElementById("name").value = "";
  document.getElementById("roll").value = "";
  document.getElementById("html").value = "";
  document.getElementById("css").value = "";
}

```

```
document.getElementById("javascript").value = "";
document.getElementById("filter").value = "All";
document.getElementById("result").innerHTML = "";
}
</script>
</body>
</html>
```

OUTPUT

Student Result

Search Student

Search

Student Name

Enter name

Roll Number

Enter roll number

Enter Marks

HTML marks

CSS marks

JavaScript marks

Filter Result

Show All

Show Result Clear

Result

Name: Aarav Sharma

Roll Number: ST101

Subject	Marks
HTML	85
CSS	78
JavaScript	90

Total: 253/300

Percentage: 84.33%

Grade: A

Result: Pass

CONCLUSION

The Interactive Student Result Dashboard successfully demonstrates how HTML, CSS, and JavaScript can be combined to create a dynamic and user-friendly academic result management system. By using JavaScript for mark calculation, grade evaluation, and result validation, the application provides quick and accurate results without requiring server-side processing.

The system effectively manages student records, subject-wise marks, percentage calculation, grade assignment, and Pass/Fail evaluation through conditional logic and JavaScript data structures. Its search and filtering features further improve usability by allowing users to quickly access specific student results.

Overall, this project demonstrates fundamental web development concepts such as DOM manipulation, event handling, conditional statements, arrays, objects, dynamic content rendering, and client-side data processing.